20 $\mathbf{a} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

(a) Write down as a column vector

(i) $\mathbf{a} + \mathbf{b}$

Add corresponding components: $\begin{bmatrix} 4 \\ 6 \end{bmatrix}$

1 mark
Column(4,6)

(1)

(ii) $2\mathbf{a} + 3\mathbf{b}$

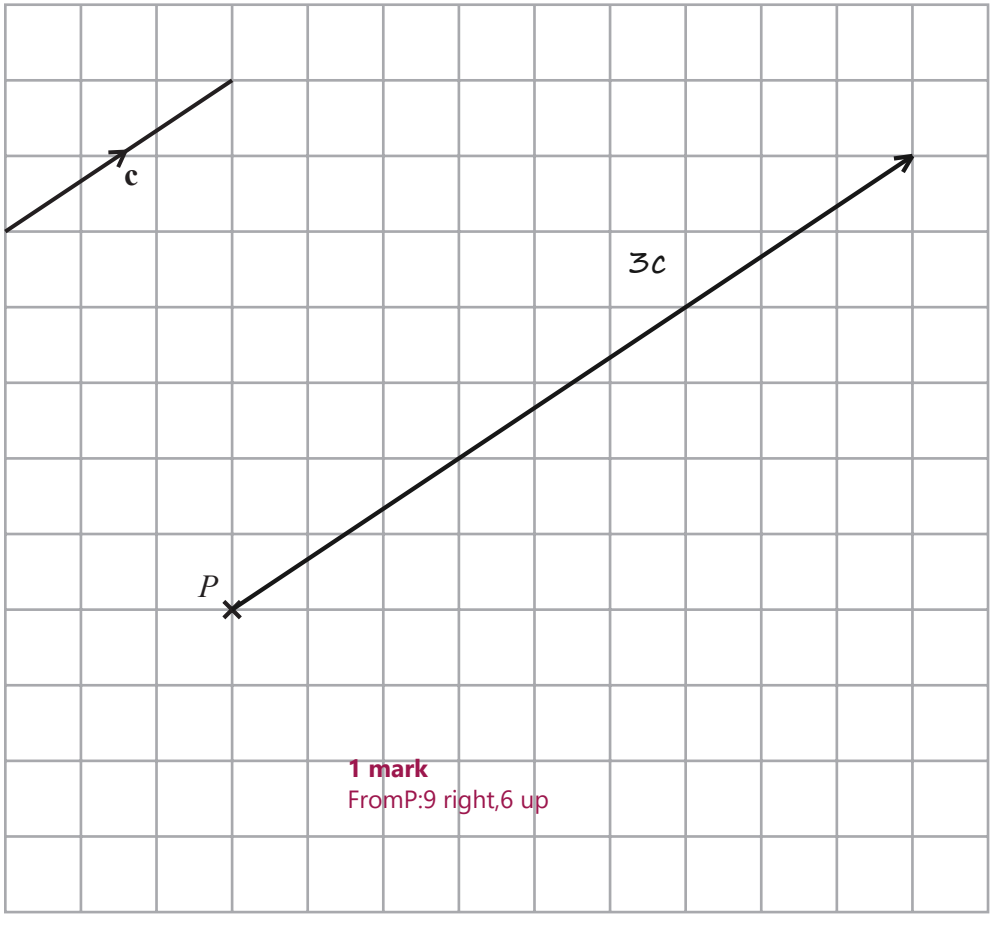
$\begin{bmatrix} 2 \\ 8 \end{bmatrix} + \begin{bmatrix} 9 \\ 6 \end{bmatrix} = \begin{bmatrix} 11 \\ 14 \end{bmatrix}$

1 mark
Scale both vectors

1 mark
Column(11,14)

(2)

The vector **c** is drawn on the grid.



1 mark
FromP:9 right,6 up

(b) From the point *P*, draw the vector 3**c**

(1)

(Total for Question 20 is 4 marks)